



CREDIT RISK LOAN

Application of Machine Learning for Credit Risk Prediction in
Costumer Loans

by M. Razy Qarar Fairuzzabadi



OUTLINE

Project Overview



Exploratory Data Analysis (EDA)



Machine Learning Modelling



Recommendation





PROJECT OVERVIEW

[Link Full Code on GitHub](#)

Background



Credit risk is one of the most important things that can affect the **stability** and **profitability** of a **financial institution**. If a **loan** is not repaid as agreed, the institution could suffer **huge losses**. Therefore, it is **crucial** for financial institutions to be able to **recognize** and **manage credit risk** well, especially in **uncertain economic conditions**.

Traditional approaches to assessing **creditworthiness**, such as **fixed rule-based credit scores**, tend to be not flexible enough to handle the complexity of various **customer data**. It also tends to be difficult to recognize **hidden patterns** or **early signs** that indicate **default risk**. Therefore, a more **modern** and **flexible approach** is needed to support a more **accurate credit scoring** process.

Machine learning provides a modern approach that enables more in-depth **data analysis** by utilizing **large amounts of historical data**. These **algorithms** can detect **patterns in customer behavior**, estimate **default probabilities**, and improve **decision-making accuracy**. By building reliable **predictive models**, financial institutions can reduce exposure to **credit risk** and strengthen **loan management systems**.

Problem Statement

1

A total of **12,29%** of the total **460 thousand customers** are still **in default**, indicating a **high level** of **credit risk** in the **loan portfolio**.

Goal

2

The **goal** of this project is to **reduce** the number of **customer defaults** by improving the **accuracy** of **credit risk assessment** using **machine learning models**.

Objective

3

Develop machine learning models to identify potentially **defaulting customers**, analyze factors that affect **credit-worthiness**, and evaluate **model performance** using relevant metrics.

Business Metrics

4

The **accuracy** of the model in classifying customers into **Good Loan** and **Bad Loan** categories is important to assess the **effectiveness** of the **credit risk prediction system**. In addition to accuracy, **Precision** is also very important. Precision measures how **precise** the model is in identifying customers who actually are in the **Bad Loan** category.

EXPLORATORY DATA ANALYSIS

EDA

[Link Full Code on GitHub](#)

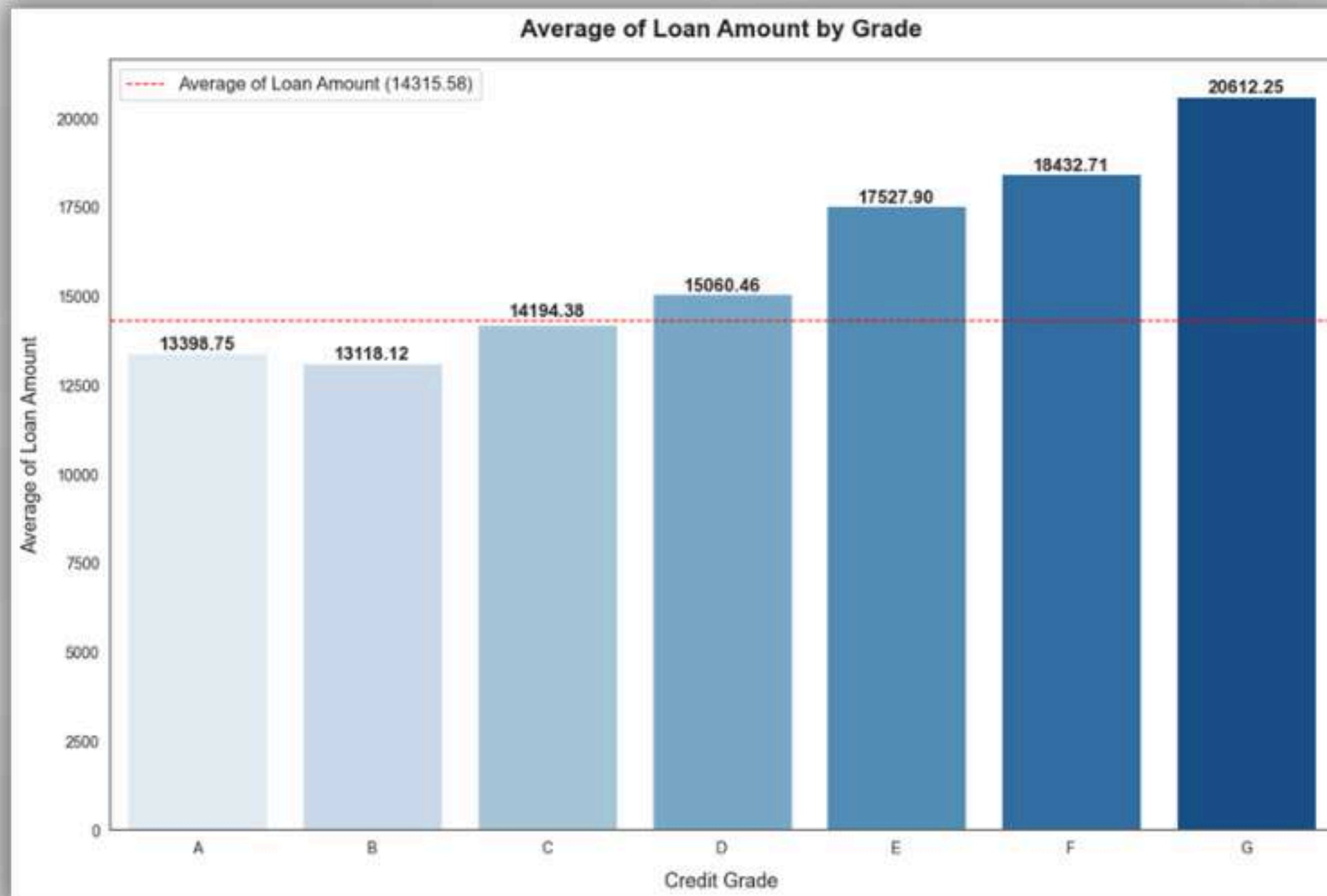




Dataset Overview

The **dataset** consists of **466285 rows** and **74 columns**, comprising **73 features** as independent variables and **1 target** variable as the dependent variable.

Analysis of Loan Amount Based on Credit Grade

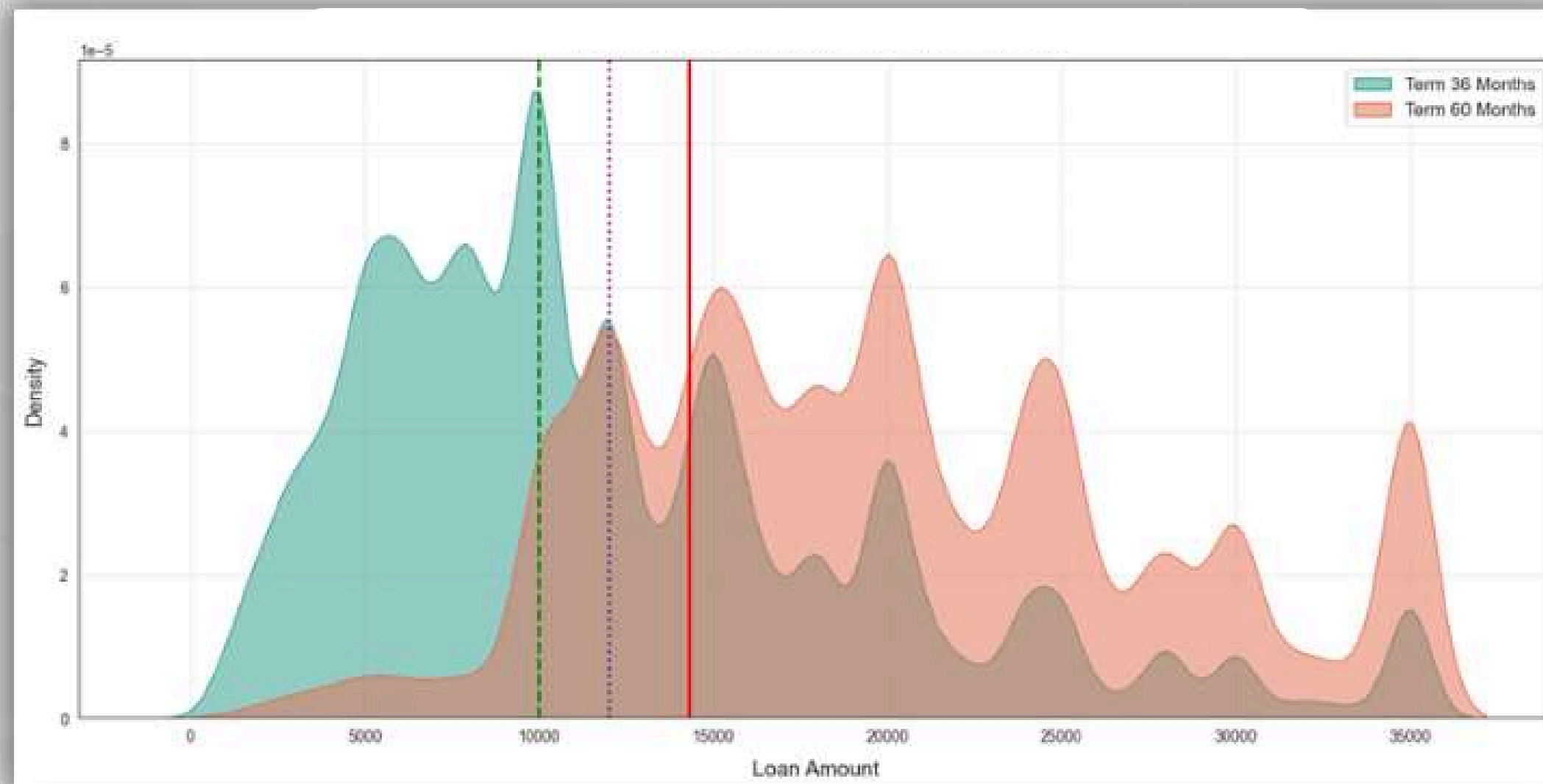


It can be seen that as the **loan level increases**, the **average loan amount** also tends to increase. For **grades A and B**, the **average loan amount** is relatively low, at around **13.398,75** and **13.118,12**. In contrast, for **grades F and G**, the **average loan amount** is much higher, at **18.432,71** and **20.612,25** respectively.

Interestingly, the line showing the **overall average loan amount (14.315,58)** is above the loan amounts in grades A and B, but below the loan amounts in higher grades, such as F and G. This suggests that an individual with a **lower credit level** receives a loan that is closer to the average, while an individual with a **higher credit level** gets a much **larger loan amount**.

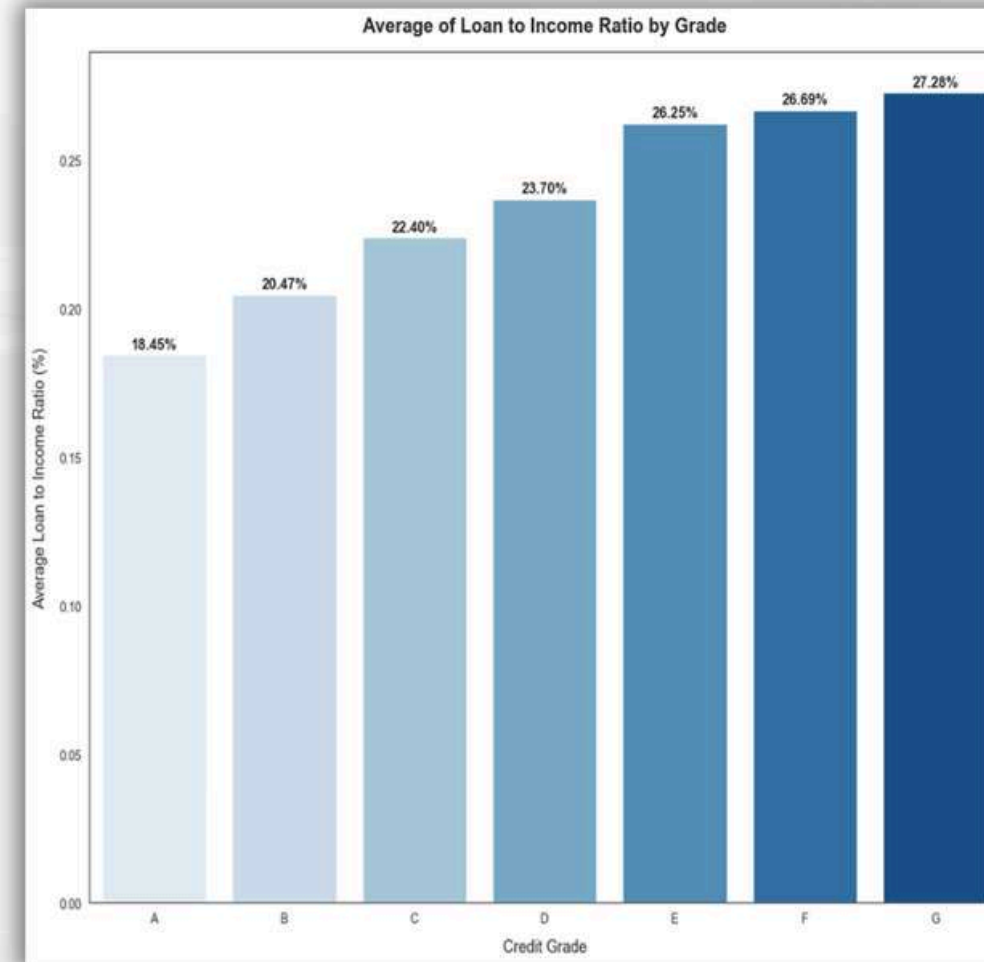
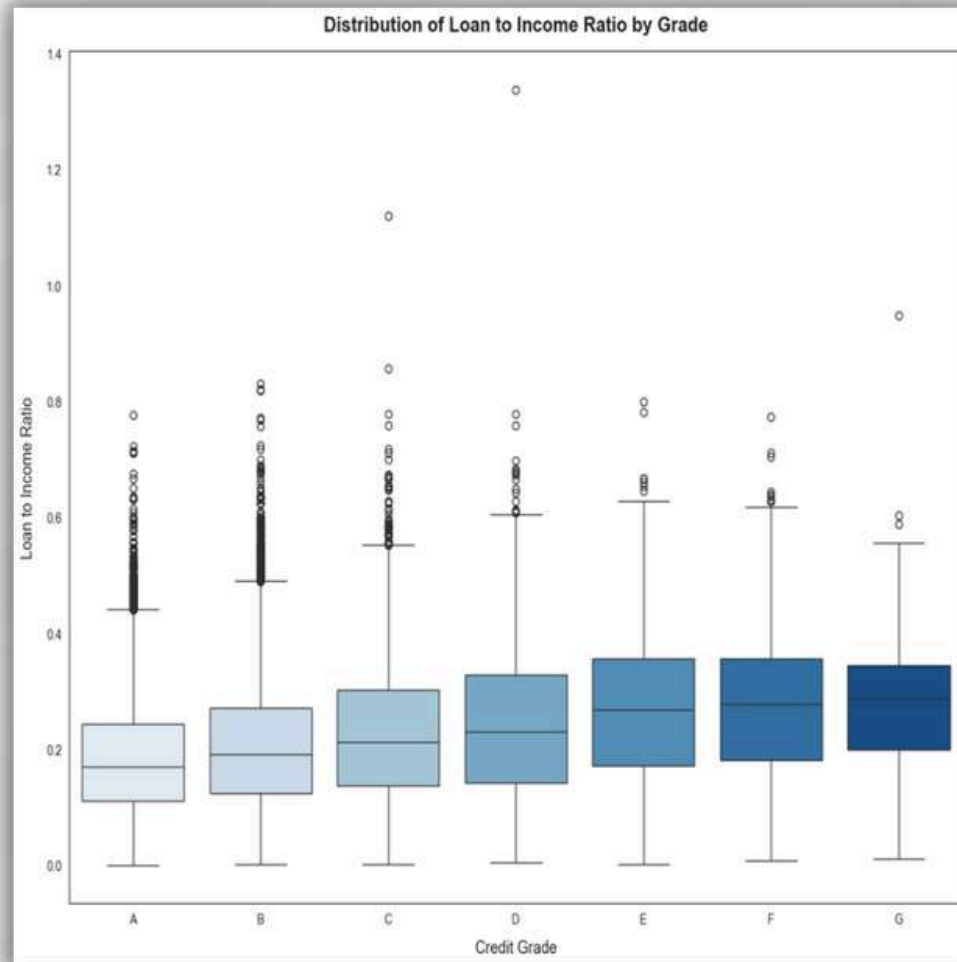
From a business perspective, this illustrates that **banking companies** or **lending companies** adjust their loan offers to the **creditworthiness** of the customer. **Larger loans** are likely to be given to an individual with a **better credit grade**, who is considered a **lower risk** to the institution.

Distribution of Loan Amount Based on Term



There is a **significant difference** between the **average loan amounts** in the **two categories**, indicating that **loans with longer tenors (60 months)** are generally granted in **larger amounts** compared to **short-term loans (36 months)**.

The **business implication** is that **banking companies** or **lending companies** need to **adjust** their **lending policies** based on **loan tenors**. Loans with **36-month tenors** tend to focus on **smaller loan amounts**, so lending companies may be more inclined to offer products with small loan amounts and short tenors to an individual who has **limited repayment capacity**. On the other hand, **60-month tenors** may be aimed at an individual who requires **larger funds**, which in turn opens up **opportunities** to offer products with **larger loan amounts** and **higher interest rates**.

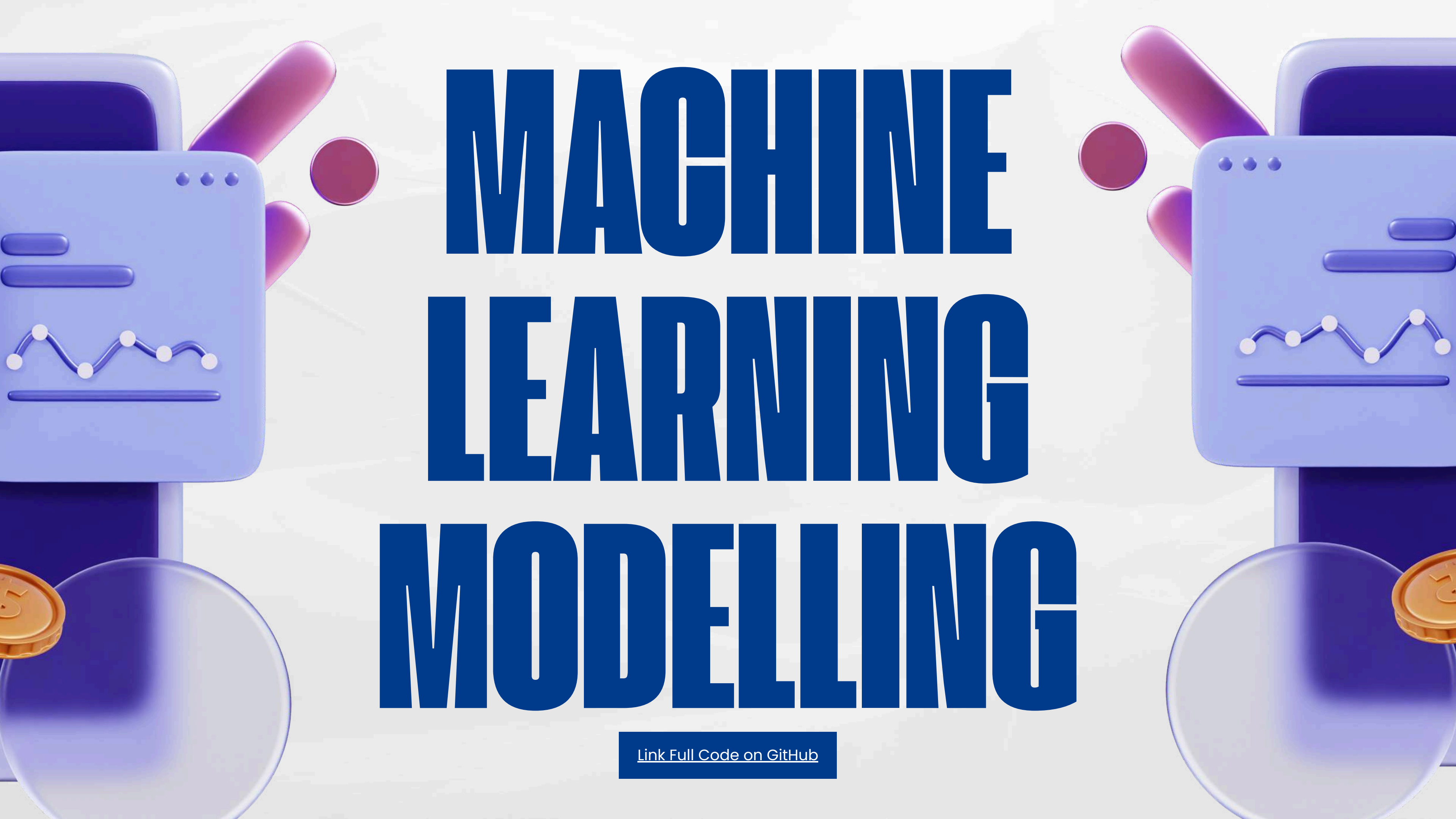


Distribution of Loan to Income Ratio by Grade & Average of Loan to Income Ratio by Grade

Based on the results of the first analysis showing the **distribution of loan-to-income ratios** by **loan grade**, it can be seen that the data provides a more detailed picture, where there are **outliers indicating the presence of an individual with very high loan ratios**, particularly at the G grade. In general, the distribution of loans tends to show that the higher the **loan grade**, the greater the **variation in loan ratios**. This indicates that **higher grades of loans** have **greater fluctuations in the proportion of loans to income** of an individual.

On the other hand, the second analysis showing the **average loan-to-income ratio** by **credit grade** shows a trend of increasing ratios as the **credit grade increases**, from grade **A** to grade **G**. Starting with **18,45% in grade A**, this ratio increases gradually until it reaches **27,28% in grade G**. This reflects that an individual with a **higher credit grade**, such as in **grade G**, tends to have a larger **loan-to-income ratio**, indicating a **higher risk** associated with their ability to make payments.

Based on these findings, **banking companies** or **lending companies** need to adjust their **lending policies** according to an **individual credit grade**. For an individual with **high credit grades**, such as **grade G**, **lending companies should consider lending smaller amounts** or **higher interest rates** to offset the **greater risk**. In addition, they should also strengthen their **risk management policies**, for example by **improving income verification** or **setting stricter requirements for loan applications** for this group. In contrast, for an individual with a **lower credit grade**, such as **grade A**, lending at a **lower ratio** may be considered to reduce the potential **risk of default**.



MACHINE LEARNING MODELLING

[Link Full Code on GitHub](#)

Data Preprocessing Workflow

1

Raw Data

The **original dataset** contains **466285 rows** and **74 columns** of data.

3

Drop Missing Values

Delete **columns** that have entirely **unique values**, are **empty**, have **constant values**, are **redundant**, or do not provide **relevant information**.

5

Filling Missing Values

Fill the **missing values** in some **columns** using the **median value**, considering that the **median** is more **robust** to **outliers** than the **average**.

2

Check Duplicated Values

There is **no duplicated data** in the **dataset**.

4

Feature Engineering

Creating **new columns**, transforming the **values** of some columns, and **adjusting data types** to suit **analysis** and **modeling needs**.

6

Pre-processing

Splitting the **data** into **training and testing sets**, **categorical feature transformation** with **Label Encoding** and **One-Hot Encoding**, **numerical feature standardization**, **relevant feature selection**, and handling **class imbalance** with **oversampling techniques** using **SMOTE**.

Modeling Workflow



1

Modeling without Hyperparameter Tuning

Running 5 machine learning algorithms, which are **Logistic Regression**, **Extra Trees Classifier**, **Random Forest**, **Adaptive Boosting**, and **Extreme Gradient Boosting**.

2

Tuning Hyperparameter

Perform **hyperparameter tuning** using **cross-validation** of the **one best performing algorithm** to optimize the modeling results.

3

Tuned Model Comparison Model Evaluation

Perform a thorough **in-person evaluation** of the selected **model** based on the **best performance**.

Base Model Comparison

Model	Accuracy	AUC	Recall	Precision	F1-Score
Logistic Regression	86,20%	82,90%	89,70%	94,20%	91,90%
Extra Trees Classifier	91,50%	82,90%	97,00%	93,40%	95,20%
Random Forest	91,80%	83,70%	97,30%	93,60%	95,40%
AdaBoost	85,30%	82,10%	88,90%	94,00%	91,40%
XGBoost	92,30%	84,20%	98,10%	93,40%	95,70%

Based on the **evaluation results** of the **five algorithms used**, **Extreme Gradient Boosting (XGBoost)** was chosen as it provided the **best performance** with an accuracy value of **92,3%** and a **precision** of **93,4%**.

Hyperparameter Tuning



	Base XGBoost				
	Accuracy	AUC	Recall	Precision	F1 – Score
Tuned XGBoost	92,30%	84,20%	98,10%	93,40%	95,70%
Base XGBoost	92,60%	84,10%	98,50%	93,40%	95,90%

Hyperparameter tuning **improved the performance of the model**, although the improvement was **not significant**.

FINAL MODEL



92,60%

Accuracy

84,10%

AUC

98,50%

Recall

93,40%

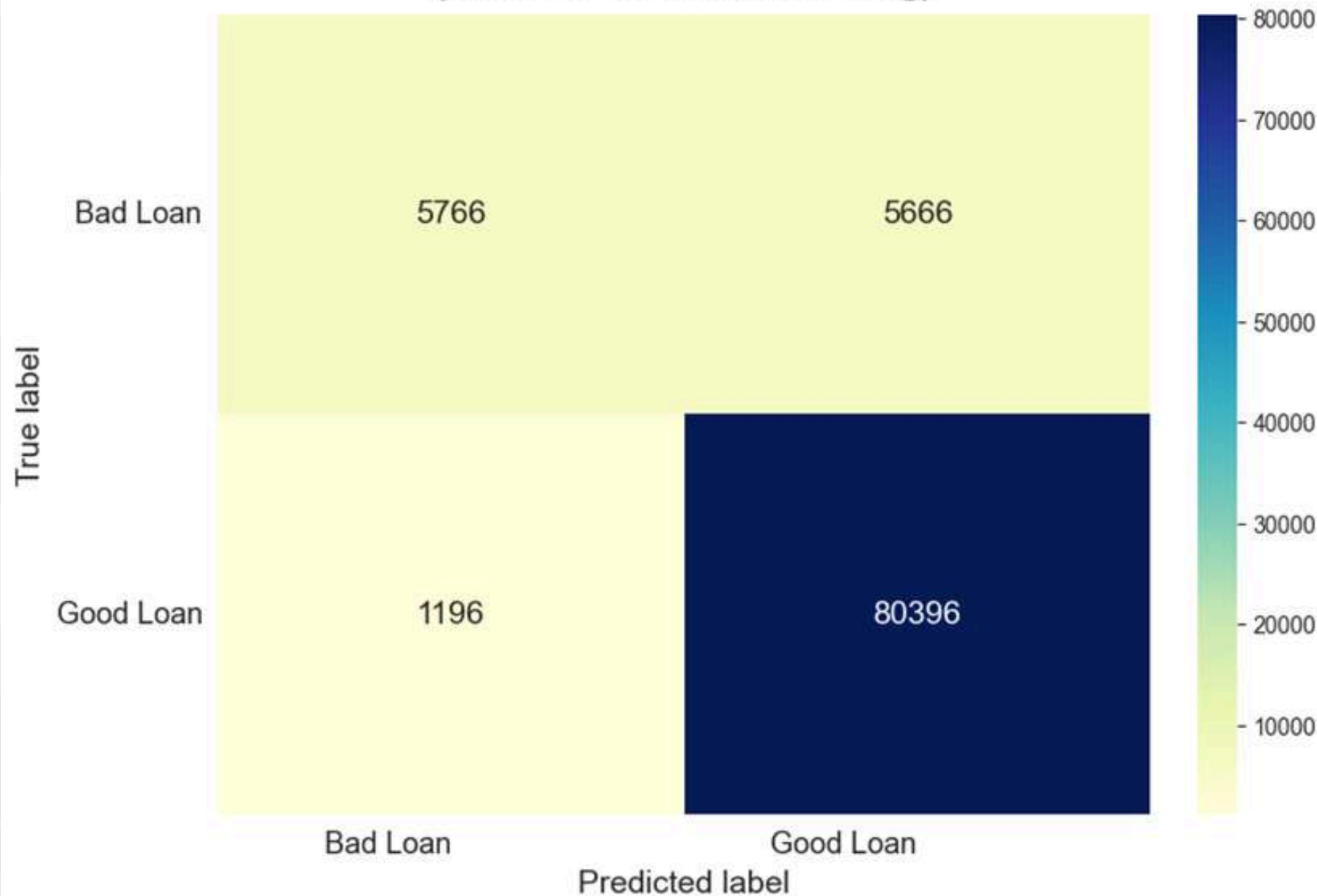
Precision

95,90%

F1 – Score

Model Evaluation

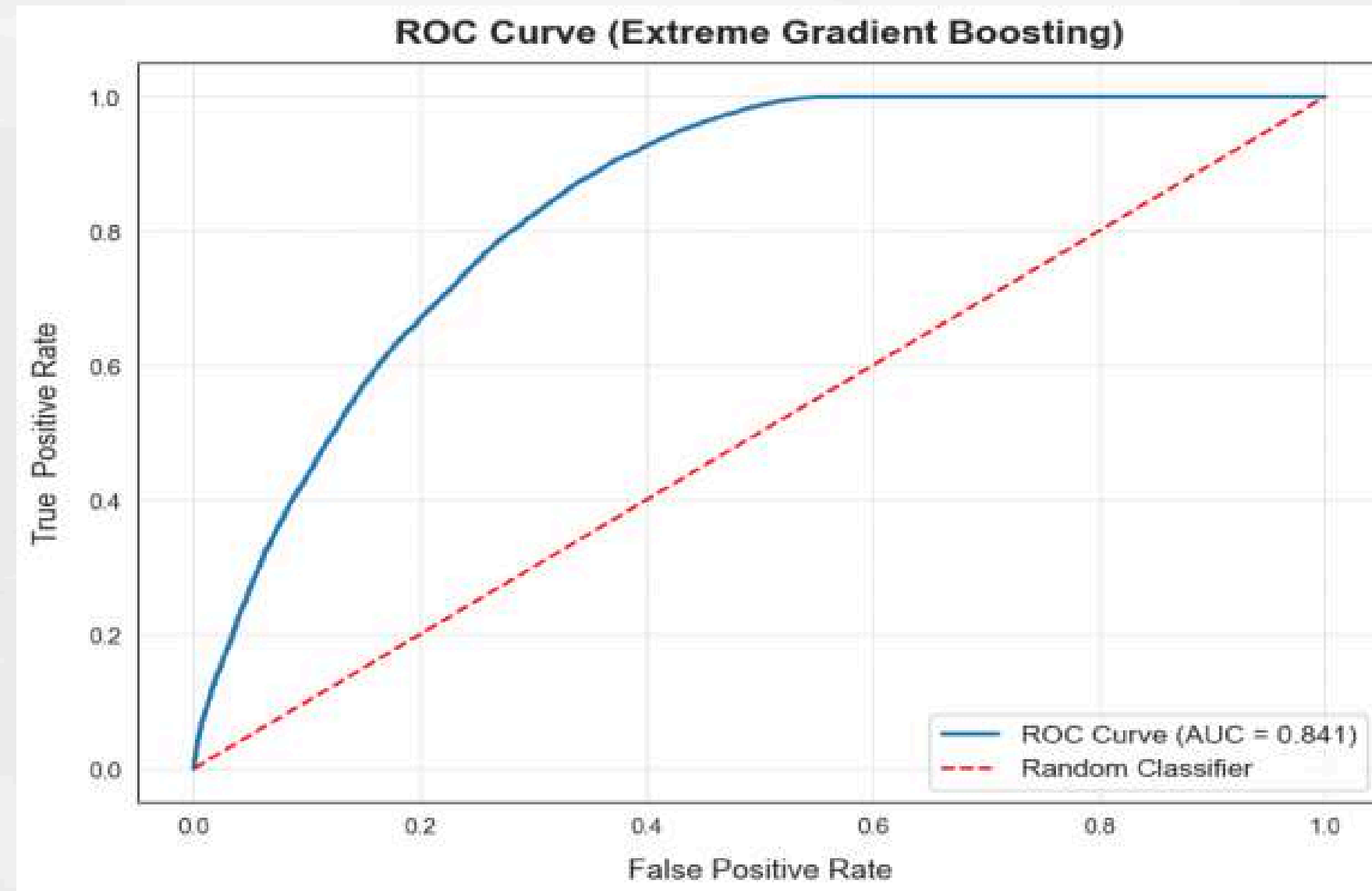
Confusion Matrix for Testing Model
(Extreme Gradient Boosting)



Based on the **confusion matrix results**, it can be seen that the model is able to **classify loans with a high success rate**, especially for the **Good Loan category**. Out of a total of **81.592 good loan data**; **80.396** were correctly classified, while **1.196** of them were **misclassified as bad loans**. In contrast, out of **11.432 bad loan data**, only **5.766** were correctly recognized, while the remaining **5.666** were classified as **good loans**.

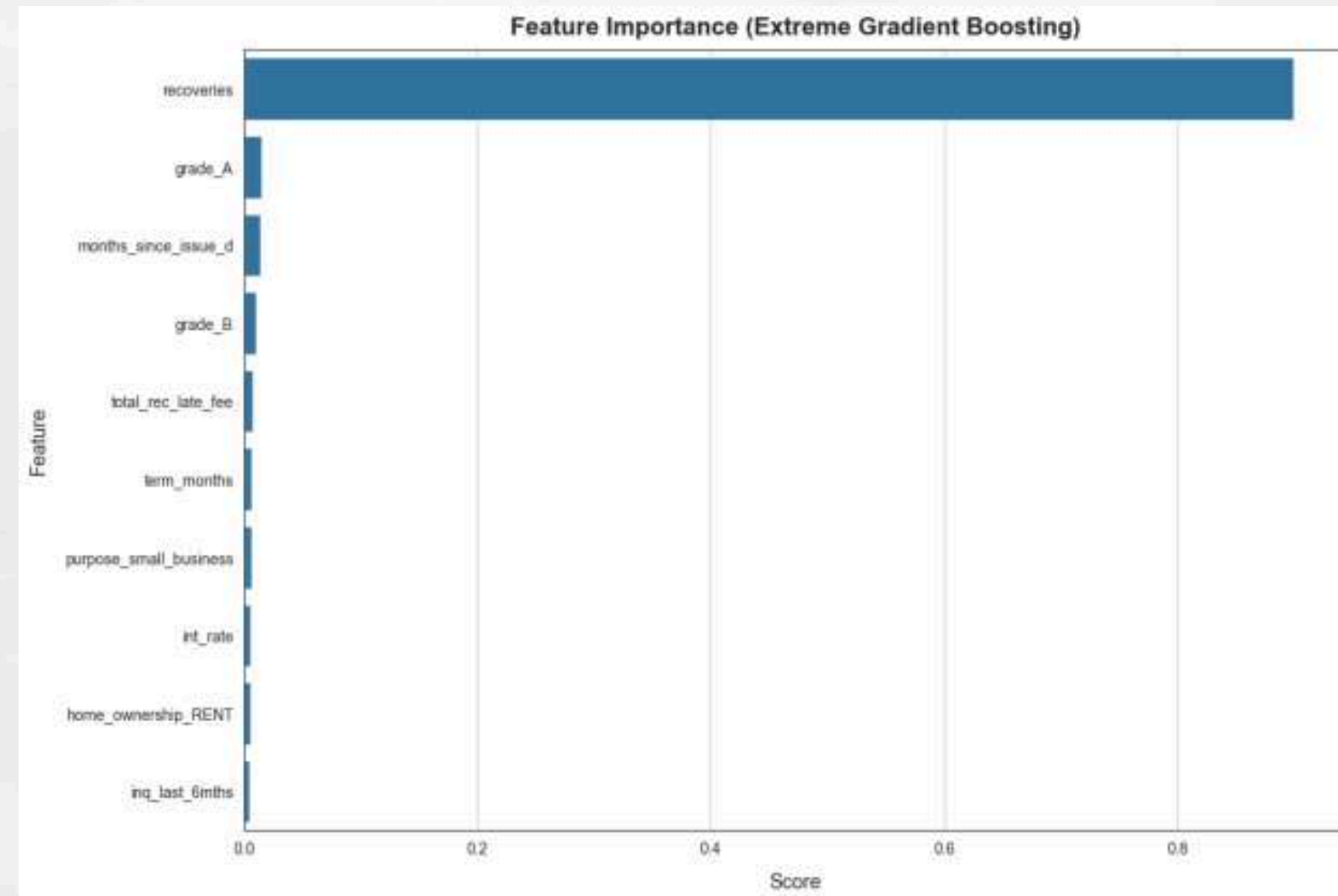
This shows that while the model is very good at recognizing loans that are classified as **secure**, it is still **less than optimal** at detecting **risky loans**. The high number of **errors on loans classified as good when they are actually bad (false positives)** is important to note as it can have a serious impact on making accurate **credit decisions**. Therefore, while the overall **performance of the model** appears promising, further improvements are needed to improve the detection of **risky loans** to make the predictions more **reliable** and **safe** to use in a **financial context**.

Model Evaluation



Based on the **ROC (Receiver Operating Characteristic) curve**, it can be seen that this model has a fairly good ability to distinguish between **Good Loan** and **Bad Loan**. This is shown by the **AUC (Area Under the Curve) value** of **0,841**, which means that the model has a probability of **84,1%** to randomly distinguish one positive example from one negative example.

Model Evaluation



Based on the **classification modeling results**, there are several **features** that have a significant influence on the borrower's loan status. The **top five features selected** are **recoveries**, **grade**, **months_since_issue_d**, **total_rec_late_fee**, and **term_months**.



RECOMMENDATION



Business Recommendation



Recoveries



recoveries refer to the **recovery** that occurs after a loan is considered to be in **default**. **Recoveries** reflect the extent to which an **unpaid loan** is recovered. To increase **recoveries**, it is important for **banking companies** or **lending companies** to focus on more effective **collections** and implement better **risk management strategies** on **high-risk loans**.



Grade



grade refer to the **loan quality category** assigned by **Lending Club** based on an assessment of the **borrower's profile**. **Banking companies** or **lending companies** should adjust **loan terms** based on this grade. **Borrowers with low grades** can be given more **competitive interest rates**, while **borrowers with high grades** should be charged **higher interest rates** or even **rejected** if they do not meet **additional requirements** such as **collateral** or **guarantees**. This strategy will help balance the **profit potential** and **default risk**.





Business Recommendation



months_since_issue_d



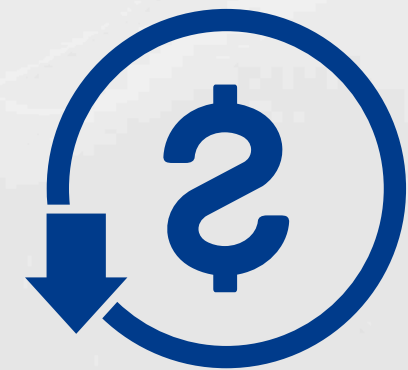
months_since_issue_d refers to the **duration (months)** after the **loan** was disbursed. This feature gives an idea of how much **time** has passed since the borrowers received the **loan funds**. **Banking companies** or **lending companies** should monitor **long-standing loans** and identify potential issues or opportunities for **restructuring loans** to make them more manageable for **long-standing borrowers**.



total_rec_late_fee



total_rec_late_fee refers to the **late fees received by the company**, reflecting the **amount of penalties imposed on borrowers** who are late with their repayments. The **higher the late fees paid by borrowers**, the more **risk they exhibit**. This information can be utilized by banking companies or lending companies to assess subsequent **creditworthiness** and categorize borrowers into appropriate **risk categories**. For **active loans**, borrowers with a history of **recurring delinquency** deserve special attention, such as through sending **automated alerts**, providing **financial counseling**, or limiting **future borrowing**.





Business Recommendation



term



term_months refers to the **number of installments in months** that the borrower must pay to repay the loan. **Banking companies** or **lending companies** being able to **customize the loan duration** to suit the borrower's ability can help improve loan repayment. For example, offering the option to **extend or shorten the loan duration** based on the loan repayment ability can help reduce the **risk of default**.



Research Advice



Use various algorithms such as SVM, K-NN, or Neural Networks.

Perform hyperparameter tuning on each model used and explore various combinations of parameters to obtain more optimal results.

Minimize false positives by considering the right trade-off between model accuracy and sensitivity.

THANK YOU

Please reach me via:

 qararfairuzzabadi@gmail.com

 <https://www.linkedin.com/in/m-razy-qarar-fairuzzabadi/>

 <https://github.com/QararFairuzzabadi>